Complete Data Analysis Masterclass

Hands-On Mentorship: Python, SQL, Power BI, Excel & Tableau

Instructor: Shubham Singh - Lead Python & Data Architecture Consultant
Industry Background: Ducat, Netmax Technologies, LNP Infotech | 550+ Students & Senior Developers Trained

BATCH SCHEDULE:	TIMINGS:	FORMAT & DURATION:
Saturday & Sunday Only	6:00 PM - 8:00 PM IST	8-10 Weeks (Live Online Hands-On)

1 Excel Advanced Modeling, Power Query & VBA

- Power Query Editor: Multi-file ingestion, unpivoting, merging/appending datasets & ETL cleanses
- VBA & Automation: Recording macros, writing subroutines, automating recurring weekly reports
- Advanced Lookup & Dynamic Arrays: XLOOKUP, INDEX/MATCH, FILTER, UNIQUE, Nested IFS
- Data Validation, Conditional Formatting & What-If Financial Analysis (Goal Seek, Scenarios)

2 SQL for Data Analysis & Relational Databases

- Relational Database Architecture, Schema Design, 3NF Normalization, Primary/Foreign Keys
- Complex Querying: Multi-Table INNER/LEFT/RIGHT/FULL JOINs, Subqueries, and Nested CTEs
- Analytical Window Functions: ROW_NUMBER, RANK, DENSE_RANK, NTILE, LEAD/LAG, Running Totals
- Performance Tuning: B-tree indexing, EXPLAIN ANALYZE, and real-world business analytics queries

3 Python Programming for Data Analytics & ETL

- Python Foundations: Data structures (Lists, Tuples, Dictionaries), Control Flow, Functions
- NumPy Computing: Vectorized array operations, matrix mathematics, indexing & slicing
- Pandas DataFrames: High-speed wrangling, missing data imputation, regex extraction, GroupBy pipelines
- Automated Data Transformation, SQLAlchemy database ingestion, and scheduled reporting scripts

4 Power BI Business Intelligence & Interactive Dashboards

- End-to-End ETL with Power Query: Cleansing, unpivoting, merging & data modeling
- Relational Data Modeling: Star Schema, Snowflake Schema, 1-to-Many Relationships
- Advanced DAX: CALCULATE, FILTER, ALL, RELATED, Time Intelligence (YTD, MTD, YoY, moving averages)
- Executive Visualization: Drill-through reports, interactive slicers, custom tooltips, KPI cards

5 Tableau Visual Storytelling & Dashboards

- Connecting to Varied Sources (Files, Relational Databases, Cloud Warehouses)
- Level of Detail (LOD) Calculations: FIXED, INCLUDE, EXCLUDE expressions & Table calculations
- Interactive Dashboards: Filter actions, highlight actions, set actions, and dynamic parameters
- Storyboarding & Cloud Publishing: Multi-chapter executive presentations on Tableau Public

6 Career Acceleration: GitHub, Resume & LinkedIn Setup

- GitHub Portfolio: Open-source repositories with production READMEs, architecture schemas & demos
- ATS-Optimized Resume: Quantified impact bullet points (Action + Tool + Metric) tailored for Data roles
- LinkedIn Setup: High-converting headline, featured projects, and recruiter direct outreach framework
- Interview Preparation: 1-on-1 mock technical interviews, live SQL problem solving on CoderPad

HANDS-ON PRODUCTION CAPSTONE PROJECTS:

1. Retail & E-Commerce Sales Performance Suite (Python ETL + SQL + Power BI Dashboard)
2. Financial Transactions & Compliance Auditing Analysis (SQL + Tableau Visual Storyboard)

All students complete live portfolio assets ready to showcase on GitHub and LinkedIn.